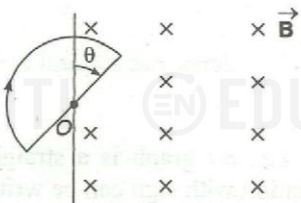


1. wire loop enclosing a semicircle of radius R is located on the boundary of uniform magnetic field B . At the moment $t = 0$, the loop is set into rotation with a constant angular acceleration α about an axis O coinciding with a line of vector $\vec{B}$ on the boundary. Find the emf induced in the loop as a function of time. Draw the approximate plot of this function. The arrow in the figure shows the emf direction taken to be positive.



Solution: The angular velocity of the loop as a function of angle θ is $\omega = \sqrt{2\alpha\theta}$

For $\theta = 0$ to π , 2π to 3π , 4π to 5π , etc., the magnetic field through the loop increases. The induced current is anticlockwise and hence emf is negative.

For $\theta = \pi$ to 2π , 3π to 5π , etc., the magnetic field through the loop decreases. The induced current is clockwise and hence emf is positive.

$$\text{As } \theta = \frac{1}{2}\alpha t^2, \quad t = \sqrt{\frac{2\theta}{\alpha}}$$

$$\text{The time taken to rotate angle } \pi \text{ is } t_1 = \sqrt{\frac{2\pi}{\alpha}}$$

$$\text{The time taken to rotate angle } 2\pi \text{ is } t_2 = \sqrt{\frac{4\pi}{\alpha}}$$

$$\text{The time taken to rotate angle } 3\pi \text{ is } t_3 = \sqrt{\frac{6\pi}{\alpha}}$$

$$\text{The time taken to rotate angle } n\pi \text{ is } t_n = \sqrt{\frac{2n\pi}{\alpha}}$$

In first half revolution, the area of loop inside the field is

$$A = \frac{1}{2}R^2\theta = \frac{1}{2}R^2\left(\frac{1}{2}\alpha t^2\right)$$

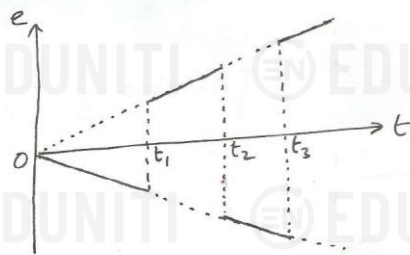
$$\text{So, flux through the loop is } \phi = BA = \frac{1}{4}BR^2\alpha t^2$$

$$\text{Hence, emf induced is } e = \left|\frac{d\phi}{dt}\right| = \frac{1}{2}BR^2\alpha t \text{ and is negative.}$$

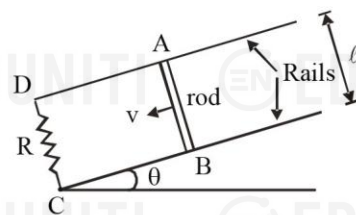
If $n = 1, 2, 3, \dots$ is the number of half revolutions performed by the loop in time t , then

$$e = (-1)^n \left(\frac{1}{2}BR^2\alpha t\right)$$

The graph of e vs t is shown in the figure.



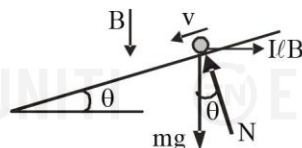
2. Two rails AD and BC are fixed parallel to each other separated by distance $\ell = AB = CD$ and inclined at angle θ with horizontal as shown in the figure.



The lower ends of the rails C and D are connected by a resistor R . A copper rod of mass m , slides down the rail under the influence of gravity. A uniform vertically downward magnetic field B exists throughout the region. Show that the rod acquires a steady-state velocity

$$v = \frac{mgR \sin \theta}{B^2 \ell^2 \cos^2 \theta}$$

Solution: The situation is shown in figure.



In steady state, the component of velocity of the rod in the horizontal direction is $v \cos \theta$ and therefore, the emf developed across it is

$$\varepsilon = B \ell v \cos \theta$$

with the positive polarity towards end B .

The current I in the rod is from A to B , i.e., directed out of the paper. It is given by

$$I = \frac{\varepsilon}{R} = \frac{B \ell v \cos \theta}{R}$$

The magnetic force acting on it is

$$F_m = I \ell B = \frac{B \ell v \cos \theta}{R} \ell B = \frac{B^2 \ell^2 v \cos \theta}{R}$$

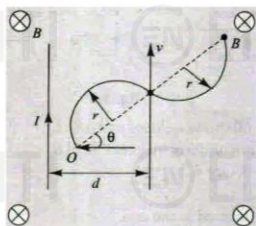
towards right as shown in the figure.

From, FBD, we have

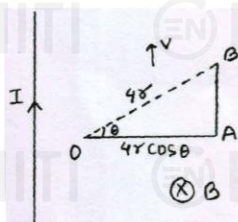
$$N \sin \theta = I \ell B = \frac{B^2 \ell^2 v \cos \theta}{R} \quad \text{and} \quad N \cos \theta = mg$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{B^2 \ell^2 v \cos \theta}{mgR} \quad \therefore v = \frac{mgR \sin \theta}{B^2 \ell^2 \cos^2 \theta}$$

3. An infinite wire carries a current I . An S-shaped conducting rod of two semicircles each of radius r is placed at an angle θ to the wire. The center of the conductor is at a distance d from the wire. If the rod translates parallel to the wire with a velocity v as shown in fig., calculate the emf induced across the ends OB of the rod.



Solution The emf induced in the S-shaped rod is same as the emf induced in the L shaped rod OAB shown in the figure.



The induced emf will be sum of the induced emfs in rods OA and AB. Induced emf in AB will be zero as the velocity of rod AB is parallel to its length. For determining emf induced in OA, consider a small element of rod of length dx at a distance x from the wire. The magnetic field due to the current in wire at the location of this element is

$$B = \mu_0 I / 2\pi x$$

The potential difference across this element is

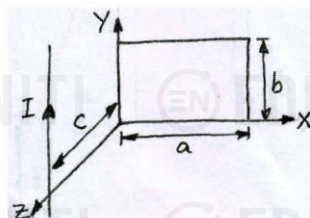
$$dV = B(dx)v = \frac{\mu_0 I v dx}{2\pi x}$$

The total potential difference is

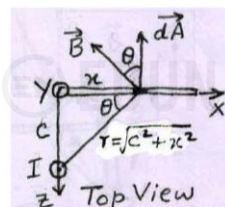
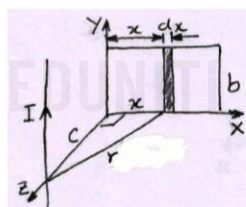
$$\Delta V = \int_{d-2r\cos\theta}^{d+2r\cos\theta} \frac{\mu_0 I v dx}{2\pi x} = \frac{\mu_0 I v}{2\pi} \ln x \Big|_{d-2r\cos\theta}^{d+2r\cos\theta}$$

$$= \frac{\mu_0 I v}{2\pi} \ln \left[\frac{d+2r\cos\theta}{d-2r\cos\theta} \right]$$

4. In fig., a long thin wire with a varying current $I = I_0 \cos \omega t$ lies at a distance c from one edge of a rectangular wire loop of length a and width b lying in the x - y plane. What emf is induced in the loop?



Solution



Consider a strip of thickness dx (of the loop) at distance x from y -axis as shown in fig. The magnetic field due to current I at the location of this strip is

$$B = \frac{\mu_0 I}{2\pi x}$$

The magnetic flux through the strip is

$$d\phi = B dA \cos \theta = \frac{\mu_0 I}{2\pi x} \times b dx \times \frac{x}{r} = \frac{\mu_0 b I x dx}{2\pi (c^2 + x^2)}$$

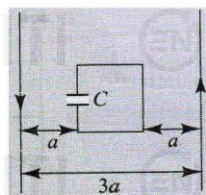
$$\Rightarrow \phi = \frac{\mu_0 b I}{4\pi} \int_0^a \frac{2x dx}{c^2 + x^2} = \frac{\mu_0 b I}{4\pi} \ln \left(\frac{c^2 + a^2}{c^2} \right)$$

where $I = I_0 \cos \omega t$

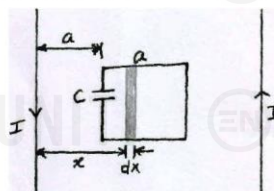
Therefore, the emf induced in the loop is

$$\varepsilon = -\frac{d\phi}{dt} = \frac{\mu_0 b I_0 \omega}{4\pi} \ln \left(\frac{c^2 + a^2}{c^2} \right) \sin \omega t$$

5. Two infinitely long parallel wires carrying currents $I = I_0 \sin \omega t$ in opposite directions are placed at a distance $3a$ apart. A square loop of side a of negligible resistance with a capacitor of capacitance C is placed in the plane of wires as shown in fig. Find the maximum current in the square loop. Also find the instantaneous charge on the upper plate of the capacitor and sketch the graph showing its variation as a function of time for one complete cycle taking anticlockwise direction of the current in the loop as positive.



Solution The magnetic flux through the square loop due to both wires is out of the page and equal in magnitude. The flux due to the left wire through the strip (of thickness dx and at distance x) shown in the figure is



$$d\phi_L = B dA = \frac{\mu_0 I}{2\pi x} a dx$$

$$\Rightarrow \phi_L = \frac{\mu_0 I a}{2\pi} \int_a^{2a} \frac{dx}{x} = \frac{\mu_0 I_0 a \ln 2}{2\pi} \sin \omega t$$

The total flux due to both wires is

$$\phi = 2\phi_L = \frac{\mu_0 I_0 a \ln 2}{\pi} \sin \omega t$$

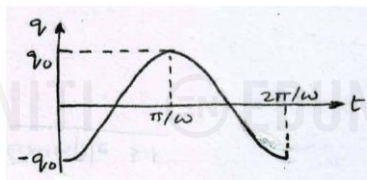
The emf induced in the loop is

$$\varepsilon = \frac{-d\phi}{dt} = \frac{-\mu_0 I_0 a \omega \ln 2}{\pi} \cos \omega t$$

Negative sign indicates that the induced emf is clockwise.

The instantaneous charge on the upper plate of capacitor is

$$q = C\varepsilon = \frac{-\mu_0 I_0 a \omega C \ln 2}{\pi} \cos \omega t$$



$$\therefore q = -q_0 \cos \omega t \text{ where } q_0 = \frac{\mu_0 I_0 a \omega C \ln 2}{\pi}$$

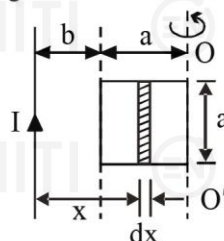
The graph of charge on the upper plate of capacitor for one complete cycle is shown in figure.

The instantaneous current in the square loop is

$$i = \frac{dq}{dt} = \frac{\mu_0 I_0 a \omega^2 C \ln 2}{\pi} \sin \omega t$$

$$\therefore i_{\max} = \frac{\mu_0 I_0 a \omega^2 C \ln 2}{\pi}$$

6. A square loop of side a and a straight infinite conductor carrying current I are placed in the same plane with two sides of the square parallel to the conductor. The inductance and resistance of the loop are equal to L and R respectively. The frame is turned through 180° about the axis OO' and stopped. Find the electric charge that flows in the square loop.



Solution: If ε is the instantaneous emf developed in the loop, then the instantaneous current i in the loop is

$$i = \frac{\varepsilon - L di/dt}{R} \Rightarrow \int R i dt = \int \varepsilon dt - \int L di$$

$$\Rightarrow Rq = \int \left(-\frac{d\phi}{dt} \right) dt - L(i_f - i_i) = (\phi_i - \phi_f) - L(i_f - i_i)$$

As the initial current i_i and the final current i_f are zero, we have

$$q = (\phi_i - \phi_f)/R$$

Now, consider a strip at a distance x in the initial position as shown in the figure. The magnetic field through this strip is along the inward normal to the plane and the flux through it is given by

$$d\phi_i = \left(\frac{\mu_0 I}{2\pi x} \right) a dx$$

$$\Rightarrow \phi_i = \frac{\mu_0 I a}{2\pi} \int_b^{a+b} \frac{dx}{x} = \frac{\mu_0 I a}{2\pi} \ln \left(\frac{a+b}{b} \right)$$

$$\text{Similarly, } \phi_f = -\frac{\mu_0 I a}{2\pi} \ln \left(\frac{2a+b}{a+b} \right)$$

The reason for the negative sign is that after rotating through 180° , the magnetic field enters the loop from opposite side.

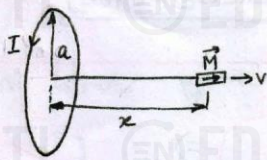
$$\Rightarrow |\phi_i - \phi_f| = \frac{\mu_0 I a}{2\pi} \ln \left(\frac{2a+b}{b} \right)$$

$$\therefore |q| = \frac{\mu_0 I a}{2\pi R} \ln \left(\frac{2a+b}{b} \right)$$

7. A small bar magnet of dipole moment $\vec{M}$ is pointing and moving with a constant velocity v along x -axis. A small closed circular conducting loop of radius a and negligible self-inductance lies in the y - z plane with its centre at $x = 0$, and its axis coinciding with the x -axis. Find the force opposing the motion of the magnet, if the resistance of the loop is R . Assume that the distance x of the magnet from the centre of the loop is much greater than a .

Solution As the magnet moves along x -axis, the magnetic flux through the loop (shown in fig.) decreases. This induces emf and hence magnetic moment in the same direction as that of the magnet. The magnetic flux through the loop due to the magnet when it is at distance x is

$$\phi = BA = \frac{\mu_0}{4\pi} \frac{2M}{x^3} \times \pi a^2 = \frac{\mu_0 M a^2}{2x^3}$$



The current in the loop due to induced emf is

$$I = \frac{\varepsilon}{R} = -\frac{1}{R} \frac{d\phi}{dt} = -\frac{1}{R} \times \frac{\mu_0 M a^2}{2} (-3x^{-4}) \frac{dx}{dt} = \frac{3\mu_0 M a^2 v}{2R x^4}$$

The magnetic moment of the loop due to this current is

$$M' = IA = \frac{3\mu_0 M a^2 v}{2R x^4} \times \pi a^2 = \frac{3\pi\mu_0 M a^4 v}{2R x^4}$$

The force opposing the motion of magnet is therefore

$$F = \frac{\mu_0}{4\pi} \frac{6MM'}{x^4} = \frac{3\mu_0 M}{2\pi x^4} \times \frac{3\pi\mu_0 M a^4 v}{2R x^4} = \frac{9\mu_0^2 M^2 a^4 v}{4 R x^8}$$

Alternatively, the magnetic field produced at the location of magnet due to the current in loop is

$$B' = \frac{\mu_0}{2} \frac{I a^2}{x^3}$$

and the magnitude of force opposing the motion of magnet is

$$F = M \left| \frac{dB'}{dx} \right| = M \left| \frac{-3\mu_0 a^2}{2x^4} I \right| = \frac{3\mu_0 M a^2}{2x^4} \times \frac{3\mu_0 M a^2 v}{2R x^4} = \frac{9\mu_0^2 M^2 a^4 v}{4 R x^8}$$

Alternate Method

As the magnet moves with a constant velocity, the force opposing its motion is equal to the force F applied by external agent towards right to maintain its velocity. The power supplied by this force is

$$P = Fv$$

This must be equal to the power dissipated in the loop.

$$\Rightarrow Fv = I^2 R = \left(\frac{3\mu_0 M a^2 v}{2R x^4} \right)^2 R \quad \therefore F = \frac{9\mu_0^2 M^2 a^4 v}{4 R x^8}$$

8. Consider a LR series circuit in which the switch is closed at $t = 0$. Let P_L be the rate at which energy increases in the inductor and P_R be the rate at which energy is dissipated as heat in the resistor.

- Express the ratio P_L / P_R as a function of time and the time constant τ of circuit.
- What is the time at which $P_L = P_R$?

Solution The current in the circuit at time t is $i = i_0 (1 - e^{-t/\tau})$

- The rate at which energy increases in the inductor is

$$P_L = \frac{dU}{dt} = \frac{d}{dt} \left(\frac{1}{2} L i^2 \right) = L i \frac{di}{dt}$$

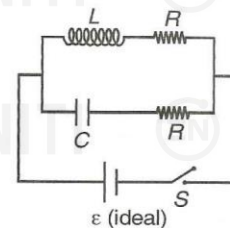
The rate at which the energy is dissipated as heat in the resistor is

$$\begin{aligned} P_R &= i^2 R \\ \therefore \frac{P_L}{P_R} &= \frac{L i \frac{di}{dt}}{i^2 R} = \frac{L}{R} \frac{di}{i} = \tau \times \frac{(i_0 / \tau) e^{-t/\tau}}{i_0 (1 - e^{-t/\tau})} \\ &= \frac{e^{-t/\tau}}{1 - e^{-t/\tau}} = \frac{1}{e^{t/\tau} - 1} \end{aligned}$$

- If $P_L = P_R$, then $\frac{1}{e^{t/\tau} - 1} = 1$

$$\Rightarrow e^{t/\tau} = 2 \quad \therefore t = \tau \ln 2$$

9. A circuit is shown in the figure. The switch is closed at $t = 0$. Find the condition for which current drawn from the battery does not vary with time.



Solution: The current through inductor and capacitor as a function of time are respectively

$$i_1 = \frac{\varepsilon}{R} (1 - e^{-Rt/L}) \quad \text{and} \quad i_2 = \frac{\varepsilon}{R} e^{-t/RC}$$

The current through battery is

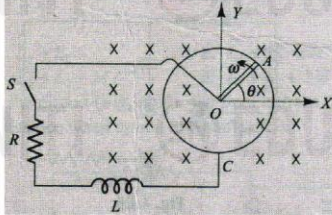
$$i = i_1 + i_2 = \frac{\varepsilon}{R} (1 + e^{-t/RC} - e^{-Rt/L})$$

As i is constant and does not vary with time, we have

$$e^{-t/RC} = e^{-Rt/L} \Rightarrow \frac{t}{RC} = \frac{Rt}{L}$$

$$\Rightarrow R^2 = \frac{L}{C} \quad \therefore R = \sqrt{\frac{L}{C}}$$

10. In the arrangement shown in fig., a metal rod OA of mass m and length r is made to rotate with a constant angular speed ω in a vertical plane about a horizontal axis at the end O . The free end A slides without friction on a fixed conducting circular ring. A uniform magnetic field B exists perpendicular to the plane of rotation. Neglect the resistance of the ring and the rod. The switch is closed at $t = 0$.



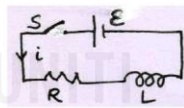
- What is the induced emf across the terminals of the switch before it is closed.
 - After the switch S is closed obtain an expression for the current as a function of time
 - When steady state is reached, obtain the time dependence of the torque required to maintain the constant angular speed.
- Consider the rod OA to be along the positive X -axis at $t = 0$

Solution

- When the switch is open, the emf across its terminals is equal to the emf induced across the rod given by

$$\varepsilon = \frac{1}{2} B \omega r^2$$

- As resistance of rod and ring is negligible the equivalent circuit can be drawn as shown in figure.



Thus, for growth of current,

$$i = \frac{\varepsilon}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \quad \therefore i = \frac{B \omega r^2}{2R} \left(1 - e^{-Rt/L} \right)$$

- The torque required to maintain constant angular speed ω is equal to the sum of torques due to magnetic force F_B on the rod and the weight mg of the rod. In steady state, $i = i_0$

$$\Rightarrow F_B = i_0 B r = \frac{B \omega r^2}{2R} \times B r = \frac{B^2 \omega r^3}{2R}$$

$$\Rightarrow \tau_B = F_B \frac{r}{2} = \frac{B^2 \omega r^4}{4R}$$

$$\tau_{mg} = mg \left(\frac{r \cos \theta}{2} \right) = \frac{mg r \cos \omega t}{2}$$

$$\therefore \tau = \tau_B + \tau_{mg} = \frac{B^2 \omega r^4}{4R} + \frac{mg r \cos \omega t}{2}$$

